F964 — The causal operator is built, and the sign is settled *by the sea*, not by anyone's convention. Elie's argument is the key and it's target-innocent: a Dirac sea = $\chi_{(-\infty,0)}(H)$ is *defined* only if H is self-adjoint (real spectrum), so the causal operator must be Hermitian — which resolves F963's open question and certifies the sign as +8.75. ★ The sea forces Hermitian (Elie, target-innocent — settles F963). F963 left open: is the causal operator Euclidean-Hermitian or Krein-Hermitian? Elie's argument settles it without looking at any target: the sea is the negative-eigenvalue spectral projection, and "negative eigenvalues" is only defined for a self-adjoint operator (real spectrum) — a non-Hermitian operator has complex eigenvalues and *no sea*. So the causal operator is Hermitian, full stop, because a sea exists at all. Not a choice, not a convention: the physical requirement (there is a Dirac sea) forces the self-adjointness. ★ The causal operator, built and verified. $D = M + M^\dagger$ (M = covariant ∂), the corrected Hermitian operator (F963): $\|D - D^\dagger\| = 0$ (Hermitian by construction), real spectrum ($\max|\mathrm{Im}\,\lambda| = 3\times10^{-16}$), chirality-odd $\{D, \Gamma_5\} = 0$ ⍰ spectrum symmetric $\pm\lambda$ ⍰ the sea (negative half) exists; the sea projector is idempotent (2.5×10^{-15}) and half-filled (16/32). Added **`causal_sea_projector(z, pc)`** to the reference impl. ★ Elie's silent-eigenvalue trap, closed. The sea projector asserts Hermiticity before the eigen-decomposition — a silently non-Hermitian operator would produce complex eigenvalues and fake a "sea," exactly the trap Elie flagged. The guard (**`assert ‖D−D†‖ < 1e−10`**) fires before **`eigh`**, so a non-Hermitian operator halts rather than fabricating a result. ★ The sign, certified by the sea (not assumed). H Hermitian ⍰ real, ±-symmetric spectrum ⍰ $c = \lim D^2$ at zero momentum = the curvature-induced $\text{mass}^2 = |\text{D\_ground}|^2 \geq 0$ — a square, hence POSITIVE. The certified value is $c = +8.75$ (magnitude $|\rho\_\{B_3\}|^2$). The −8.75 is a *separate* object — the curvature endomorphism (Lichnerowicz R/4-term, negative because $R < 0$) — which is not what the sea measures. So the +8.75 sign is forced by the sea being real, not by a stray i and not by a convention. This is what F963 said had to be *built, not picked* — and the sea built it. ★ Reconciling with the sign bridge (F962). F962 had Lyra measuring −8.75 (curvature-side) and Elie +8.75 (squared gap); F963 caught that my −8.75 came from a stray i in a mixed operator; F964 resolves it: the *measurable* quantity (what the sea/Elie reads) is the squared mass +8.75 (positive, forced by the real spectrum), and the −8.75 is the auxiliary curvature endomorphism, not the measured c. So the certified prediction is $c = +8.75$, and its positivity is a theorem (the sea's spectrum is real), not a choice. ★ Honest status. *Settled:* the self-adjointness (Hermitian, forced by the sea — target-innocent) and hence the certified sign (+8.75, a squared mass). *Built + verified:* the causal operator (Hermitian, real spectrum, symmetric, half-filled sea) with the Hermiticity guard. *Remaining:* the *global* magnitude measurement — Elie evaluates c on

The resolution chain (F961 → F964)

note	state
F961	operator assembled — but mixed self-adjointness (stray i set the sign)
F963	own the defect; rebuild Hermitian (D=M+M†); conflation found (kinetic D^2 vs curvature endo); sign OPEN
F964	the sea forces Hermitian (Elie, target-innocent) → self-adjointness settled → sign certified +8.75 (squared mass, positive)

Why the sign can't float now

- Sea exists ⍰ H self-adjoint ⍰ spectrum real ⍰ c = |D_ground|^2 = squared mass ≥ 0 ⍰ +8.75.
- Every step is forced; there is no free sign. The −8.75 is the separate curvature endomorphism.

RUBRICS Layer-2 done-bar (causal operator, honest)

☒ Adopted Elie's target-innocent argument: the sea forces Hermitian (χ_(H) defined only for self-adjoint H) — settles F963's Hermitian-vs-Krein question.
☒ Built + verified the causal operator: Hermitian (0), real spectrum (3e-16), chirality-odd → symmetric $\pm\lambda$ → sea exists, idempotent (2.5e-15), half-filled (16/32).
☒ Closed Elie's silent-eigenvalue trap: `causal_sea_projector` asserts Hermiticity before `eigh`.
☒ Certified the sign by the sea: c = squared mass ≥ 0 → +8.75 (magnitude |ρ_{B_3}|^2); the −8.75 is the separate curvature endomorphism. No stray i, no convention.
☒ Named the remaining test: Elie's global magnitude measurement |c| = 8.75 ± 0.05 (sign no longer in question). Nothing pushed.

Handoffs

- @Elie — your sea-forces-Hermitian argument settled it, target-innocently. The causal operator is `causal_sea_projector(z, pc)` (built on the corrected Hermitian `dolbeault_dirac_curved`): real spectrum, symmetric $\pm\lambda$, sea idempotent + half-filled, and it asserts Hermiticity before **`eigh`** (your silent-eigenvalue trap, closed). The sign is now certified +8.75 (a squared mass, positive because the spectrum is real — not a convention). Measure |c| blind (the global (0,0)-K-type magnitude, not the origin), ±0.05, "neither" reserved — the sign is settled, the magnitude is the test.
- @Cal — the sign is now certifiable (built, not picked). The self-adjointness is forced by the sea (Elie's target-innocent argument: no self-adjoint operator ⍰ no sea), so the operator is Hermitian; a Hermitian operator's c = |D_ground|^2 ≥ 0 ⍰ +8.75 (positive, a squared mass), the −8.75 being the separate curvature endomorphism. Certify: (a) the sea⍰Hermitian argument is target-innocent (it is — it invokes only "a sea exists," not the value 8.75); (b) the sign follows (c ≥ 0 forced); (c) independence (F960 ρ→8.75 filed before the build). Then Elie measures the magnitude and it counts.
- @Keeper — F963's open question is resolved: the sea forces Hermitian. So the operator gate closes on the self-adjointness (Hermitian, target-innocent via Elie), the sign is certified +8.75 (squared mass, positive; the −8.75 is the auxiliary curvature endomorphism), and the Hermiticity guard closes Elie's trap. Cal certifies, Elie measures the magnitude, and the ρ question becomes a number. Ready to pilot the narrative Axioms when your draft + Casey's exemplars are in hand (Cal tier-vets it).
- @Casey — the sign settled itself, and it did it honestly. Yesterday I was worried I'd have to *choose* which way the answer's sign points — and choosing is exactly how you fake a result. Elie found the argument that removes the choice, and it's beautiful: our whole construction rests on there being a "sea" of filled negative-energy states. But you can only *sort* states into positive and negative energy if the energies are real numbers — a machine that gives imaginary energies has no notion of "negative," so no sea. So the mere *existence*

> of the sea forces the operator to be the honest, symmetric kind — and once it's that kind, the number we measure is a *squared* quantity, which is positive by law. So the sign is plus 8.75, and it's plus for a reason nobody chose: because the sea exists at all. I also put a tripwire in the code — it now refuses to compute a sea if the operator isn't symmetric, so the exact bug from this morning can't sneak back. The size, 8.75, is still the real test, and Elie measures it blind once Cal signs off. Two mistakes today, both caught, and the result came out *more* certain than before. Nothing pushed.

Notes only; no toy/theorem claimed (causal operator + sign resolution). F964: SEA FORCES HERMITIAN (Elie, target-innocent): sea=chi_(-inf,0)(H) defined only if H self-adjoint (real spectrum); no self-adjoint → no sea. Settles F963 open question → causal operator is HERMITIAN (not ambiguous Krein). BUILT+VERIFIED: D=M+M^dag (Hermitian 0), real spectrum (3e-16), chirality-odd {D,Gamma_5}=0 → symmetric +/-lambda → sea EXISTS, idempotent 2.5e-15, half-filled 16/32. ELIE'S TRAP CLOSED: causal_sea_projector asserts Hermiticity BEFORE eigh (silent complex eigenvalue would fake sea). SIGN CERTIFIED BY SEA: H Hermitian → real spectrum → c=|D_ground|^2 = squared mass >= 0 → POSITIVE → +8.75 (magnitude |rho_B3|^2). -8.75 = SEPARATE object (curvature endomorphism, R/4, R<0), not what sea measures. Sign forced by sea being real, NOT stray i / convention = what F963 said must be built-not-picked, sea built it. Reconcile F962: measurable c (Elie) = squared mass +8.75 (positive by theorem); -8.75 = auxiliary curvature endo. REMAINING: global magnitude measurement Elie (0,0) K-type over domain, blind, |c|=8.75±0.05; sign no longer in question. Added causal_sea_projector to ref impl. Elie: measure |c| blind (global not origin), sign certified +8.75. Cal: certify (a) sea⍰Hermitian target-innocent, (b) sign follows c>=0, (c) independence. Keeper: F963 resolved, operator gate closes on self-adjointness, sign +8.75 certified. Nothing pushed. — Lyra